

FEU Domain KT — Concepts & Note-Taking Scaffold

EY Probation Training — Week 3 (FEU Domain KT) and parts of Week 4.

Read this differently from the other six documents. Everything else here is knowledge you can acquire independently. Week 3 is **internal domain knowledge** that exists only in FEU's SharePoint, their KT sessions, and their people's heads. No external source can supply it.

What this document does: gives you the **generic supply-chain forecasting vocabulary** so the sessions land on prepared ground, and provides a **structured set of questions** to fill in as you go. Treat the blanks as the actual deliverable.

1. Generic Demand-Forecasting Vocabulary

These concepts are industry-standard. FEU's implementation will differ in detail, but the underlying ideas are stable.

Planning hierarchy

Demand forecasting almost always operates over two intersecting hierarchies:

Product hierarchy — e.g. Item / SKU → Brand → Sub-category → Category → Total. **Location or customer hierarchy** — e.g. Store → Customer → Channel → Region → Total.

Forecasts are generated at one level and **aggregated up** or **disaggregated down**.

- **Top-down** — forecast the aggregate, split it using historical proportions. Stable but loses item-level signal.
- **Bottom-up** — forecast each item, sum upward. Captures detail but noisy at the item level.
- **Middle-out** — forecast at an intermediate level, then go both directions.
- **Reconciliation** — making the levels mathematically consistent after independent forecasts.

The general principle: aggregation reduces noise, so accuracy measured at a higher level always looks better. This is why the **level of measurement must always be stated**.

GTIN

Global Trade Item Number — the GS1 standard product identifier, the number behind a barcode. Variants: GTIN-8, GTIN-12 (UPC), GTIN-13 (EAN), GTIN-14 (case/carton level).

Important for forecasting because the same physical product can carry different GTINs across pack sizes and markets, and GTINs get retired and reissued. Any join on GTIN needs care about which level and which time period.

APG

Not a universal standard acronym — in planning contexts it most often means an **Aggregated Planning Group** or **Article Planning Group**: a grouping of products planned together, typically because they share demand behaviour, substitute for each other, or are too sparse to forecast individually.

Confirm FEU's exact definition. This is precisely the kind of term where assuming the generic meaning will make you sound confidently wrong.

Customer hierarchy

The buying-side structure — ship-to location → sold-to customer → customer group → channel → region. Retailers often have deep hierarchies where forecasts are agreed at a negotiated level.

Baseline vs total forecast

- **Baseline** — the underlying demand pattern excluding promotions, events and one-offs. Usually what the statistical model produces.
- **Uplift** — the incremental demand attributable to a promotion or event.
- **Total / final forecast** — baseline + uplift + manual overrides.

The distinction matters because baseline accuracy and final accuracy are different measurements, and a team can be good at one and poor at the other.

Forecast lifecycle

A typical cycle, which FEU's will resemble: 1. **Data ingestion** — sales history, master data, promotions calendar, price. 2. **Cleansing** — outlier treatment, removing the effect of stockouts (censored demand), history alignment for product transitions. 3. **Baseline generation** — statistical or ML model. 4. **Enrichment** — promotional uplift, new-product forecasting, judgemental overrides. 5. **Review / consensus** — demand planners adjust; sometimes a formal S&OP meeting. 6. **Publication** — the forecast is locked and passed to supply planning. 7. **Measurement** — accuracy and bias computed once actuals arrive, at agreed lags.

Lag and horizon

- **Horizon** — how far ahead you forecast (e.g. 13 weeks).
- **Lag** — how far in advance the forecast being measured was made. "Lag-1 accuracy" measures the forecast made one period before; "lag-3" measures the one made three periods before.

Accuracy always degrades with lag. **Comparing accuracy figures at different lags is meaningless**, and quoting a lag-1 number as if it were a lag-3 number is a classic way to overstate performance.

Other terms you may meet

- **Intermittent / lumpy demand** — many zero periods; standard metrics like MAPE break down. Croston's method and its variants exist for this.
- **Cannibalisation** — a new or promoted product taking demand from another.
- **Halo effect** — a promotion lifting demand for related products.
- **Phase-in / phase-out** — new product introduction and discontinuation; history must be linked across the transition.
- **Censored demand** — sales don't equal demand when you were out of stock. Uncorrected, this teaches the model that demand was lower than it was.
- **S&OP / IBP** — Sales & Operations Planning / Integrated Business Planning; the governance process forecasts feed into.

2. Forecasting KPIs — Recap

Covered fully in the ML & Forecasting document; repeated here because Week 3 revisits them from the business side.

- **Forecast Accuracy (FA)** — typically $(1 - \text{WMAPE}) \times 100\%$.
- **Forecast Bias (FB)** — **signed**; positive is over-forecasting, negative is under-forecasting.
- **WMAPE** — $\sum |A-F| / \sum A$; the volume-weighted workhorse.
- **Tracking signal** — cumulative bias over MAD; flags systematic drift.

Report **accuracy and bias together, always**. Accuracy alone hides systematic over- or under-forecasting, and bias is what actually accumulates into excess stock or stockouts.

The three questions to ask about any KPI figure quoted at FEU: 1. At what **hierarchy level** is it measured? 2. At what **lag**? 3. Is it **baseline** or **final** forecast?

Without those three, an accuracy number means nothing — and asking them makes you look like you understand the domain.

3. Note-Taking Scaffold

Fill these in during KT sessions. Blank entries are gaps to chase.

Business context

- What does FEU stand for, and what is its scope?
- Which business or client does it serve, and what decisions consume its output?
- What is the cost of a bad forecast here — stockouts, waste, working capital, service penalties?
- Who are the consumers of the forecast, and what do they do with it?
- What is the planning cadence (weekly? monthly?) and the cut-off timing?

Data

- What are the source systems, and who owns each?
- What is the grain of the input data (item $\times$ location $\times$ day?), and the forecast output grain?
- How much history is available and used?
- What master data is involved, and what are the known quality issues?
- How are promotions, prices and calendars supplied?

Hierarchies

- Exact product hierarchy levels, top to bottom
- Exact customer/location hierarchy levels
- **FEU's definition of APG** — write it down verbatim
- At which level is the forecast generated? Aggregated? Agreed with the business?
- Top-down, bottom-up or middle-out?

Modelling

- What models are in production, and why those?
- What is the baseline definition, and how is uplift computed?
- What features are used?
- How and when is the model retrained?
- How are new products and discontinuations handled?
- How is intermittent demand handled?

KPIs

- **FEU's exact FA formula** — write out the numerator and denominator
- **FEU's exact FB formula** — and the sign convention
- At what level and lag are they reported?
- What are the current accuracy and bias figures, and the targets?
- Who reviews them, and how often?

Technical

- End-to-end flow from source to published forecast
- Which components run in ADF vs Databricks
- Storage layout and zone names

- Where the code lives; repo structure and branch policy
- How a change reaches production
- Where jobs are monitored and how failures are alerted

Operations

- The backtesting process — how it's run, what it reports, what "good" looks like
- Parallel run — duration, comparison metrics, exit criteria
- Common production failures and their runbooks
- Support rotation and escalation path

4. How to Use the KT Sessions Well

The sessions are the only source for the above, and Week 4's end-of-week activity says explicitly that the doubt-clearing session is meant to leave you "ready for final assessment."

- **Take notes in this structure**, not linearly. You'll immediately see which cells are empty.
- **Ask about definitions, not just process.** "How is FA calculated here?" is a better question than "what's your accuracy?" — and the answer is more useful.
- **Ask for one worked example end to end** — a single SKU through the whole pipeline. One concrete trace teaches more than three architecture diagrams.
- **Write down acronyms immediately**, with the speaker's own words. Every organisation has terms that mean something slightly non-standard.
- **Save the doubt-clearing sessions for genuine gaps**, not for things in the SharePoint docs — read those first.
- **Note who owns what.** Knowing whom to ask is itself the deliverable in a support context.

5. Assessment Preparation

Week 3 and 4 expected outcomes are all explanation-shaped: *able to explain FEU E2E flow, able to explain forecasting terminology used in FEU, able to explain hierarchy structures, understand FEU production support workflow.*

So prepare to **explain out loud**, not to recall definitions:

- ☐ Describe the FEU end-to-end flow in two minutes, source to consumption
- ☐ Draw the architecture from memory, naming each component's role
- ☐ Explain the product and customer hierarchies and at which level forecasting happens
- ☐ Define FA and FB using FEU's formulas, and say why bias matters
- ☐ Walk through what happens when a production run fails
- ☐ Explain what backtesting and parallel run each verify
- ☐ Trace one SKU from raw sales through to published forecast

If you can do all seven without notes, Weeks 3 and 4 are covered.